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DS-B

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HOMEWORK #13

1. $\langle p,q \rangle = p(x_0)q(x_0) + p(x_1)q(x_1) + p(x_2)q(x_2) + p(x_3)q(x_3)$

$p = p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$

$q = q(x) = b_0 + b_1x + b_2x^2 + b_3x^3$

@ ~~$x_1 = 1$~~ $x_0 = -3$, $x_1 = -1$, $x_2 = 1$, and $x_3 = 3$

(a) $p(x) = x^2 \mapsto \mathbb{P}_1 = \text{Span}\{1, x\}$

Let $f(x) = a + bx$ in $\mathbb{P}_1$

$f(x) = 1 + x \longmapsto f_1 = 1$, $f_2 = x$

$$\text{proj}_{\mathbb{P}_1} p = \frac{\langle p, f_1 \rangle}{\langle f_1, f_1 \rangle} f_1 + \frac{\langle p, f_2 \rangle}{\langle f_2, f_2 \rangle} f_2$$

$$\frac{\langle p, f_1 \rangle}{\langle f_1, f_1 \rangle} f_1 = \frac{p(x_0)f_1(x_0) + p(x_1)f(x_1) + p(x_2)f_1(x_2) + p(x_3)f_1(x_3)}{f_1(x_0)^2 + f_2(x_1)^2 + f_2(x_2)^2 + f(x_3)^2} f_1$$

$$= \frac{9 + 1 + 1 + 9}{1 + 1 + 1 + 1}(1) = \frac{20}{4}(1) = 5$$

$$\frac{\langle p, f_2 \rangle}{\langle f_2, f_2 \rangle} f_2 = \frac{p(x_0)f_2(x_0) + p(x_1)f_2(x_1) + p(x_2)f_2(x_2) + p(x_3)f(x_3)}{f_2(x_0)^2 + f_2(x_1)^2 + f_2(x_2)^2 + f_2(x_3)^2} f_2$$

$$= \frac{(9)(-3) + (1)(-1) + (1)(1) + (9)(3)}{9 + 1 + 1 + 9} x = 0$$

$\text{proj}_{\mathbb{P}_1} p = 5 + 0 = 5$ Ans

$p_{approx}(x) = 5$

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(b) $\mathbb{P}_2 = \text{Span}\{1, x, x^2\}$

$f_1(x) = 1$, $f_2(x) = x$ $f_3(x) = x^2$

$u_1 = f_1 \mapsto$ orthogonal basis $= \{u_1, u_2, u_3\}$

$u_2 = f_2 - \text{proj}_{u_1} f_2 = x - \frac{\langle u_1, f_2 \rangle}{\langle f \rangle} \frac{\langle f_2, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1$

known from (a)

$= x - \frac{\langle x, 1 \rangle (1)}{\langle 1, 1 \rangle} = x - 0 = x$

$u_3 = f_3 - \text{proj}_{u_1} f_3 - \text{proj}_{u_2} f_3$

$u_3 = x^2 - \frac{\langle x^2, 1 \rangle (1)}{\langle 1, 1 \rangle} - \frac{\langle x^2, x \rangle x}{\langle x, x \rangle}$

(a)

$u_3 = x^2 - 5 - 0 = x^2 - 5$

Orthogonal Basis $= \{1, x, x^2 - 5\}$

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2. $\langle A, B \rangle = tr(A^T B)$, $\beta = \{ \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{M_1}, \underbrace{\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}}_{M_2}, \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}}_{M_3}, \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}}_{M_4} \}$ Ahmed.

Orthogonal $\beta = \{u_1, u_2, u_3, u_4\}$

Orthonormal $\beta = \{e_1, e_2, e_3, e_4\}$

$$e_1 = \frac{u_1}{\|u_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$e_2 = M_2 - proj_{e_1} M_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} - 1 \langle A_2, e_1 \rangle e_1$$

$$u_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} - tr(A_2^T e_1) \frac{1}{\sqrt{2}} tr \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$e_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$u_3 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} - proj_{e_1} M_3 - proj_{e_2} M_3$$

$$u_3 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} - \langle M_3, e_1 \rangle e_1 - \langle M_3, e_2 \rangle e_2$$

$$u_3 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} - tr(A_3^T e_1) e_1 - tr(A_3^T e_2) e_2$$

$$u_3 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 0 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$e_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

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($u_4 = M_4 - \text{proj}_{e_1} M_4 - \text{proj}_{e_2} M_4 - \text{proj}_{e_3} M_4$ Hmm

$u_4 = M_4 - \langle M_4, e_1 \rangle e_1 - \langle M_4, e_2 \rangle e_2 - \langle M_4, e_3 \rangle e_3$

$u_4 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} - \begin{bmatrix} 0 \end{bmatrix} - \begin{bmatrix} 0 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

$e_4 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

$E = \left\{ \begin{bmatrix} 1/\sqrt{2} & 0 \\ 0 & 1/\sqrt{2} \end{bmatrix}, \begin{bmatrix} 0 & -1/\sqrt{2} \\ 1/\sqrt{2} & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1/\sqrt{2} \\ 1/\sqrt{2} & 0 \end{bmatrix}, \begin{bmatrix} 1/\sqrt{2} & 0 \\ 0 & -1/\sqrt{2} \end{bmatrix} \right\}$

3. $f(x) = x - 1 \quad , [-\pi, \pi]$

$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x - 1 \; dx = \frac{1}{2\pi} \left| \frac{x^2}{2} - x \right|_{-\pi}^{\pi} = -1$

$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} (x-1) \cos kx \; dx = \frac{1}{\pi} \int_{-\pi}^{\pi} x \cos x \; dx - \frac{1}{\pi} \int_{-\pi}^{\pi} \cos x \; dx$

$a_k = 0$ as odd function

Odd functio so change limits

$a_1 = \frac{2}{\pi} \left[\int_0^{\pi} x \cos x \; dx - \int_0^{\pi} \cos x \; dx \right]$

$a_2 = -\frac{4}{\pi}$

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$$a_k = \frac{1}{\pi}\int_{-\pi}^{\pi}(x-1)\cos(kx)\,dx$$

$$a_k = \frac{1}{\pi}\left[\int_{-\pi}^{\pi} x\cos kx\,dx - \int_{-\pi}^{\pi}\cos kx\,dx\right]$$

$a_k = 0$ because symmetric interval on odd function

$$b_k = \frac{1}{\pi}\int_{-\pi}^{\pi}(x-1)\sin kx\,dx$$

$$b_k = \frac{1}{\pi}\left[\int_{-\pi}^{\pi} x\sin kx\,dx - \int_{-\pi}^{\pi}\sin kx\,dx\right]$$

$$b_k = -\frac{2\pi\cos k\pi}{\pi k}$$

Fourier Approximation: $-1 + \sum_{k=1}^{n}\left(-\frac{2\pi\cos k\pi}{k}\right)$